Continuum Mechanics

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Chapter 01

Tensor Algebra

Part 2 of 3

Section 1.3
Second-Order Tensors

Second-Order Tensors

Definition

A *second-order tensor* on the vector space $\mathcal{V}$ is a linear mapping

$$T : \mathcal{V} \longrightarrow \mathcal{V}.$$

Thus,

$$\begin{aligned} T(\mathbf{u}+\mathbf{v}) &= T\mathbf{u} + T\mathbf{v}, & \mathbf{u},\mathbf{v} &\in \mathcal{V}, \\ T(\alpha\mathbf{u}) &= \alpha T\mathbf{u}, & \alpha \in \mathbb{R}, \quad \mathbf{u} &\in \mathcal{V}. \end{aligned}$$

The *zero tensor* O and *identity tensor* I satisfy

$$O\mathbf{v} = \mathbf{0}, \qquad I\mathbf{v} = \mathbf{v}, \qquad \mathbf{v} \in \mathcal{V}.$$

Two tensors T and S are equal if $T\mathbf{v} = S\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.

Example (Projection)

Suppose that $\mathbf{e} \in \mathcal{V}$ is a unit vector:

$$\mathbf{e} \cdot \mathbf{e} = 1.$$

Define, for every $\mathbf{v} \in \mathcal{V}$,

$$P\mathbf{v} = (\mathbf{v} \cdot \mathbf{e})\mathbf{e}.$$

Then P is the projection onto $\mathbf{e}$, and P is a second-order tensor.

Example (Rotation)

Suppose we wish to rotate every $\mathbf{v} \in \mathcal{V}$ through an angle θ about the fixed axis $\mathbf{e}_3$, where

$$\mathcal{F} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

is a given frame. Define

$$\begin{aligned} S\mathbf{e}_1 &= \cos(\theta)\mathbf{e}_1 + \sin(\theta)\mathbf{e}_2, \\ S\mathbf{e}_2 &= \cos(\theta)\mathbf{e}_2 - \sin(\theta)\mathbf{e}_1, \\ S\mathbf{e}_3 &= \mathbf{e}_3. \end{aligned}$$

These values completely determine the second-order tensor S, because every vector is a linear combination of the frame vectors.

Example (Reflection)

Suppose M is the unique plane through the origin with unit normal $\mathbf{e}$:

$$\mathbf{e} \cdot \mathbf{e} = 1, \qquad \mathbf{v} \cdot \mathbf{e} = 0 \quad \text{for every } \mathbf{v} \in M.$$

Set

$$T\mathbf{v} = \mathbf{v} - 2(\mathbf{v} \cdot \mathbf{e})\mathbf{e}.$$

Then T is a second-order tensor that reflects vectors across M.

What is $T(T\mathbf{v})$?

Exercise

Create a second-order tensor that projects the image of $\mathbf{v}$ *onto the plane* M*. What are the properties of this tensor?*

Tensor Algebra

Addition and Scalar Multiplication

Definition

The set of all second-order tensors is denoted $\mathcal{V}^2$. Suppose $S, T \in \mathcal{V}^2$ and $\alpha \in \mathbb{R}$. Then $S + T$ and αT are defined by

$$(S + T)(\mathbf{v}) = S\mathbf{v} + T\mathbf{v},$$
$$(\alpha T)(\mathbf{v}) = \alpha(T\mathbf{v}),$$

for every $\mathbf{v} \in \mathcal{V}$.

It is not hard to show that $S + T \in \mathcal{V}^2$ and $\alpha T \in \mathcal{V}^2$.

Tensor Products by Composition

Definition

Suppose that $S, T \in \mathcal{V}^2$. The *product* of S and T is the tensor ST defined by

$$(ST)(\mathbf{v}) = S(T\mathbf{v}), \qquad \mathbf{v} \in \mathcal{V}.$$

One can show directly that $ST \in \mathcal{V}^2$.

Consequently, $\mathcal{V}^2$ is a vector space and an algebra.

Representation in a Given Frame

Components of a Tensor

Definition

Suppose that $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame and $S \in \mathcal{V}^2$. The coordinates, or components, of S are the nine scalars

$$S_{ij} = \mathbf{e}_i \cdot (S\mathbf{e}_j), \qquad 1 \leq i, j \leq 3.$$

Proposition

Suppose $\mathcal{F} = \{\mathbf{e}_i\}$ *is a frame,* $\mathbf{v} = v_i \mathbf{e}_i \in \mathcal{V}$*, and* $S \in \mathcal{V}^2$*. Set*

$$\mathbf{u} := S\mathbf{v}.$$

Then the coordinates u_i *of* $\mathbf{u}$ *in* $\mathcal{F}$ *satisfy*

$$u_i = S_{ij} v_j, \qquad 1 \leq i \leq 3, \tag{1.6}$$

where summation over the dummy index j *is understood and* i *is free.*

Proof.

By definition,

$$u_m \mathbf{e}_m = \mathbf{u} = S(\mathbf{v}) = S(v_n \mathbf{e}_n).$$

Since S is linear,

$$\begin{aligned} u_m \mathbf{e}_m &= S(v_n \mathbf{e}_n) \\ &= v_n S \mathbf{e}_n. \end{aligned}$$

Take the dot product with $\mathbf{e}_k$:

$$\mathbf{e}_k \cdot (u_m \mathbf{e}_m) = \mathbf{e}_k \cdot (v_n S \mathbf{e}_n).$$

Proof (Cont.)

Since the dot product is linear and the frame is orthonormal,

$$\begin{aligned}\mathbf{e}_k \cdot (u_m \mathbf{e}_m) &= u_m(\mathbf{e}_k \cdot \mathbf{e}_m) \\ &= u_m \delta_{km} \\ &= u_k .\end{aligned}$$

Similarly,

$$\begin{aligned}\mathbf{e}_k \cdot (v_n S \mathbf{e}_n) &= v_n \mathbf{e}_k \cdot (S \mathbf{e}_n) \\ &= v_n S_{kn} \\ &= S_{kn} v_n .\end{aligned}$$

Thus $u_k = S_{kn} v_n$, which is (1.6) after renaming the free index and dummy index as i and j. □

Component and Matrix Notation

When we want to emphasize the ij component of S with respect to the frame $\mathcal{F} = \{\mathbf{e}_i\}$, we write

$$[S]_{ij} = S_{ij} = \mathbf{e}_i \cdot (S\mathbf{e}_j).$$

We also write S as the matrix

$$[S] = \begin{pmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{pmatrix} \in \mathbb{R}^{3\times 3}.$$

The component $[S]_{ij}$ lies in the ith row and jth column.

Transpose of a Tensor Matrix

Definition

Let $S \in \mathcal{V}^2$, and let $[S]$ be its matrix representation in the frame $\mathcal{F}$. The transpose $[S]^T$ is

$$[S]^T = \begin{pmatrix} S_{11} & S_{21} & S_{31} \\ S_{12} & S_{22} & S_{32} \\ S_{13} & S_{23} & S_{33} \end{pmatrix}.$$

In other words,

$$[S]_{ij}^T = [S]_{ji}.$$

Exercise

Let $S \in \mathcal{V}^2$ *and suppose* $\mathcal{F} = \{\mathbf{e}_i\}$ *is a given frame. Show that*

$$[S] = \begin{bmatrix} | & | & | \\ [S\mathbf{e}_1] & [S\mathbf{e}_2] & [S\mathbf{e}_3] \\ | & | & | \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

That is, the j*th column of* $[S]$ *is the coordinate column vector* $[S\mathbf{e}_j]$*.*

If $\mathbf{u} = S\mathbf{v}$*, conclude that*

$$[\mathbf{u}] = [S][\mathbf{v}],$$

the usual matrix-vector multiplication from linear algebra.

Example (Rotation)

For the rotation tensor introduced earlier,

$$
\begin{aligned}
S\mathbf{e}_1 &= \cos(\theta)\mathbf{e}_1 + \sin(\theta)\mathbf{e}_2,\\
S\mathbf{e}_2 &= -\sin(\theta)\mathbf{e}_1 + \cos(\theta)\mathbf{e}_2,\\
S\mathbf{e}_3 &= \mathbf{e}_3.
\end{aligned}
$$

Thus its matrix in the frame $\mathcal{F} = \{\mathbf{e}_i\}$ is

$$
[S] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.
$$

Example (Identity Tensor)

Consider the identity tensor $I\mathbf{v} = \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$. Let $\mathcal{F} = \{\mathbf{e}_i\}$ be any orthonormal frame. Then

$$\begin{aligned} I_{ij} &= \mathbf{e}_i \cdot (I\mathbf{e}_j) \\ &= \mathbf{e}_i \cdot \mathbf{e}_j \\ &= \delta_{ij}. \end{aligned}$$

Thus the components of I are $I_{ij} = \delta_{ij}$.

Dyadic Products

The Dyadic Product

Definition

Let $\mathbf{a}, \mathbf{b} \in \mathcal{V}$ be given. The second-order tensor $\mathbf{a} \otimes \mathbf{b}$, defined by

$$(\mathbf{a} \otimes \mathbf{b})\mathbf{v} = (\mathbf{b} \cdot \mathbf{v})\mathbf{a}, \qquad \mathbf{v} \in \mathcal{V},$$

is called the *dyadic product* of $\mathbf{a}$ and $\mathbf{b}$.

Proposition

Suppose $\mathcal{F} = \{\mathbf{e}_i\}$ *is a frame and*

$$\mathbf{a} = a_i \mathbf{e}_i, \qquad \mathbf{b} = b_i \mathbf{e}_i.$$

Then

$$[\mathbf{a} \otimes \mathbf{b}]_{ij} = a_i b_j.$$

Here i *and* j *are free indices.*

Proof.

By definition,

$$
\begin{aligned}
[\mathbf{a} \otimes \mathbf{b}]_{ij} &= \mathbf{e}_i \cdot \big((\mathbf{a} \otimes \mathbf{b}) \mathbf{e}_j \big) \\
&= \mathbf{e}_i \cdot \big((\mathbf{b} \cdot \mathbf{e}_j) \mathbf{a} \big) \\
&= \mathbf{e}_i \cdot (b_j \mathbf{a}) \\
&= b_j (\mathbf{e}_i \cdot \mathbf{a}) \\
&= b_j a_i \\
&= a_i b_j .
\end{aligned}
$$

$\square$

Exercise

Show that, for a given frame $\mathcal{F} = \{\mathbf{e}_i\}$,

$$[\mathbf{a} \otimes \mathbf{b}] = [\mathbf{a}][\mathbf{b}]^T.$$

The transpose turns the 3×1 *coordinate column* $[\mathbf{b}]$ *into a* 1×3 *row, so the product is a* 3×3 *matrix.*

Proposition

Let $S \in \mathcal{V}^2$ be given and suppose $\mathcal{F} = \{\mathbf{e}_i\}$ is any frame. Define, as usual,

$$S_{ij} = [S]_{ij} = \mathbf{e}_i \cdot (S\mathbf{e}_j).$$

Then

$$S = S_{ij}(\mathbf{e}_i \otimes \mathbf{e}_j), \tag{1.7}$$

where both i and j are dummy indices.

Proof.

Let $\mathbf{u} \in \mathcal{V}$ be arbitrary and set

$$\mathbf{v} := S\mathbf{u}.$$

Write

$$\mathbf{v} = v_m \mathbf{e}_m, \qquad \mathbf{u} = u_r \mathbf{e}_r.$$

Then

$$\begin{aligned} S_{ij}(\mathbf{e}_i \otimes \mathbf{e}_j)\mathbf{u} &= S_{ij}(\mathbf{e}_i \otimes \mathbf{e}_j)(u_r \mathbf{e}_r) \\ &= S_{ij} u_r \big((\mathbf{e}_i \otimes \mathbf{e}_j)\mathbf{e}_r\big). \end{aligned}$$

Proof (Cont.)

Using the definition of the dyadic product,

$$\begin{aligned} S_{ij}u_r\big((\mathbf{e}_i \otimes \mathbf{e}_j)\mathbf{e}_r\big) &= S_{ij}u_r\big((\mathbf{e}_j \cdot \mathbf{e}_r)\mathbf{e}_i\big) \\ &= S_{ij}u_r(\delta_{jr}\mathbf{e}_i) \\ &= S_{ij}u_j\mathbf{e}_i \\ &= v_i\mathbf{e}_i \\ &= S\mathbf{u}, \end{aligned}$$

where equation (1.6) gives $v_i = S_{ij}u_j$. Since $\mathbf{u}$ is arbitrary,

$$S = S_{ij}(\mathbf{e}_i \otimes \mathbf{e}_j).$$

□

Second-Order Tensor Algebra in Components

Addition and Scalar Multiplication in Components

If $S, T \in \mathcal{V}^2$ and $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame, then

$$[S + T] = [S] + [T]$$

and

$$[\alpha T] = \alpha[T].$$

Here $[\,\cdot\,]$ denotes the matrix representation with respect to $\mathcal{F}$.

Exercise

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame. Suppose $S, T \in \mathcal{V}^2$, and set $U = ST$. Show that

$$U_{ij} = S_{ik} T_{kj}, \qquad 1 \leq i, j \leq 3,$$

where i, j are free and k is dummy.

To prove this result, we first establish the following dyadic-product identity.

Proposition

For any $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathcal{V}$,

$$(\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d}) = (\mathbf{b} \cdot \mathbf{c})\, \mathbf{a} \otimes \mathbf{d}.$$

Proof.

Let $\mathbf{v} \in \mathcal{V}$ be arbitrary. Then

$$\begin{aligned}
\big((\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d})\big)\mathbf{v} &= (\mathbf{a} \otimes \mathbf{b})\big((\mathbf{c} \otimes \mathbf{d})\mathbf{v}\big) \\
&= (\mathbf{a} \otimes \mathbf{b})\big((\mathbf{d} \cdot \mathbf{v})\mathbf{c}\big) \\
&= (\mathbf{d} \cdot \mathbf{v})(\mathbf{a} \otimes \mathbf{b})\mathbf{c} \\
&= (\mathbf{d} \cdot \mathbf{v})(\mathbf{b} \cdot \mathbf{c})\mathbf{a}.
\end{aligned}$$

Proof (Cont.)

Continuing,

$$
\begin{aligned}
(\mathbf{d}\cdot\mathbf{v})(\mathbf{b}\cdot\mathbf{c})\mathbf{a} &= (\mathbf{b}\cdot\mathbf{c})\big((\mathbf{d}\cdot\mathbf{v})\mathbf{a}\big)\\
&= (\mathbf{b}\cdot\mathbf{c})\big((\mathbf{a}\otimes\mathbf{d})\mathbf{v}\big)\\
&= \big((\mathbf{b}\cdot\mathbf{c})(\mathbf{a}\otimes\mathbf{d})\big)\mathbf{v}.
\end{aligned}
$$

Since $\mathbf{v}$ is arbitrary,

$$
(\mathbf{a}\otimes\mathbf{b})(\mathbf{c}\otimes\mathbf{d}) = (\mathbf{b}\cdot\mathbf{c})\,\mathbf{a}\otimes\mathbf{d}.
$$

□

Solution of the Exercise.

Using (1.7) and the preceding proposition,

$$\begin{aligned} U &= (S_{ij}\mathbf{e}_i \otimes \mathbf{e}_j)(T_{k\ell}\mathbf{e}_k \otimes \mathbf{e}_\ell) \\ &= S_{ij}T_{k\ell}(\mathbf{e}_i \otimes \mathbf{e}_j)(\mathbf{e}_k \otimes \mathbf{e}_\ell) \\ &= S_{ij}T_{k\ell}(\mathbf{e}_j \cdot \mathbf{e}_k)(\mathbf{e}_i \otimes \mathbf{e}_\ell) \\ &= S_{ij}T_{k\ell}\delta_{jk}(\mathbf{e}_i \otimes \mathbf{e}_\ell) \\ &= S_{ik}T_{k\ell}(\mathbf{e}_i \otimes \mathbf{e}_\ell). \end{aligned}$$

Thus

$$U_{i\ell} = S_{ik}T_{k\ell}.$$

$\square$